ONEMODO PATRONS

Smart Gate Control & Quarry Logistics System

Complete Operational Workflow, Hardware Integration & API Technical Manual

Application: Onemodo Patrons (Mobile App)	**Target Environment:** Android & iOS
Core Framework: React Native & Expo Router	**State Manager:** Redux Toolkit
Server Host: https://sat.modomines.com/apiv3/	**Date of Documentation:** October 2026

1. System Overview & Objectives

The **Onemodo Patrons** mobile application is designed specifically for quarry gate security personnel, weighbridge operators, and fleet supervisors. Its primary mission is to automate vehicle ingress/egress, eliminate unauthorized gate access, and maintain 100% accurate time-in-quarry tracking for both third-party visitor trucks and internal company-owned fleets.

Key Operational Capabilities:

- **Dual-Step Automated Gate Access:** Instant photo verification followed by barcode ticket scan.
- **Direct LAN Boom Barrier Triggering:** Communicates with IoT relay controllers via local network HTTP calls to open and close security barriers in real-time.
- **Autocomplete & Searchable Registry:** Pre-fetches registered commercial trucks to eliminate manual data entry errors.
- **Real-Time Pending Vehicle Monitor:** Live dashboard of all active vehicles inside the quarry with live turnaround timers.
- **Automated Daily Exit Logging:** Daily filtered departures for auditing and material reconciliation.
- **Employee Self-Service:** Shift punch attendance records and digital PDF payslip viewing.

2. System Architecture & Authentication

The application architecture consists of three interconnected layers:

Mobile App Client	**Cloud Backend APIs**	**Gate Relay Hardware**
React Native / Expo	https://sat.modomines.com/apiv3/	Local IP Controller (LAN)
• Redux Auth Store	• JWT Authentication	• Relay 1: Gate Close
• Camera & Barcode Engine	• Truck Plate Validation	• Relay 2: Gate Open
• Dynamic Segmented UI	• Trip Logs & Barcode DB	• 5000 Port HTTP Switch

JWT Security & Dynamic Company Routing:

When an employee logs in via /employeeLogin, the server returns a cryptographically signed JWT. The app decodes the token payload on the device to extract employee_id and company_id (e.g., Company 23). Every subsequent vehicle entry, barrier configuration request, and log query is dynamically scoped to that exact company.

3. Core Operational Workflows

3.1 Vehicle Gate Entry Flow (Register Entry & Open Gate)

1. **Select Wheeler Count:** The operator selects the truck category (e.g., 6 Wheeler, 10 Wheeler, 12 Wheeler, Car, Bike).
2. **Search / Enter Plate:** The operator types the registration plate. The auto-complete dropdown searches /activevehiclelist in real time. If matched, the vehicle type is auto-selected.
3. **Photo Capture:** The operator captures a live photo of the incoming vehicle using the integrated camera.
4. **Submission:** Tapping **'Register Entry & Open Gate'** sends the vehicle data to /boomentry.
5. **Barrier Open:** Upon API success, the app immediately fires an HTTP command to the local barrier controller (Relay 2) to lift the barrier.
6. **Status:** The vehicle moves to the *'Vehicles Inside Quarry'* active list.

Important Validation Rule: For Visitor vehicles, the is_own parameter is strictly omitted so the backend defaults to visitor without triggering validation errors. For Own Fleet vehicles, is_own: 1 is passed.

3.2 Quick-Step Exit Flow (Photo -> Barcode Scan -> Gate Open)

To maximize weighbridge throughput, the exit process is engineered into an automated 2-step pipeline:
• **Step 1 — Immediate Exit Photo:** The operator taps **'Scan & Exit'** on any vehicle card inside the quarry. The camera immediately launches to photograph the departing truck and cargo.
• **Step 2 — Automated Barcode Scanner:** As soon as the exit photo is taken, the screen seamlessly switches to the high-speed barcode scanner.
• **Step 3 — Auto Exit & Barrier Lift:** The operator scans the ticket barcode (e.g., 3354841). The app instantly calls /boomexit (passing only barcode and exit_image) and sends an automated command to the gate controller to lift the boom barrier.
• **Fallback Override:** For non-truck vehicles (cars, motorcycles) or damaged tickets, a **'Skip Barcode & Open Gate'** option performs a direct exit (barcode: 0).

3.3 Logs & Traffic Monitoring

• **Inside Quarry (Pending List):** Lists all vehicles currently inside the premises. Displays duration spent on-site (e.g., 45m), entry timestamp, wheeler type, and entry photo preview.
• **Today's Exited Records:** Automatically queries /boomexitedlist for today's date (from_date = today, to_date = today) without requiring manual date picker selection.

4. Complete API Endpoint Reference

All APIs use Base URL: https://sat.modomines.com/apiv3/. All endpoints accept User-Agent: DashboardApp.

#	Method & Endpoint	Category	Key Request Parameters	Expected Output
1	**POST** `employeeLogin`	Auth	username, password	JWT Token, employee details, shift hour
2	**POST** `boomsettings`	Hardware	company_id, token	Local relay URLs for barrier open & close
3	**GET** `activevehiclelist`	Lookup	company_id, search	Array of registered trucks & wheelers
4	**POST** `boomentry`	Gate Entry	truck, vehicle_type, entry_datetime, entry_image, company_id, [is_own:1]	Saves entry record, generates ticket barcode
5	**GET** `boompendinglist`	Gate Logs	company_id, truck, limit, offset	Active vehicles currently inside quarry
6	**POST** `boomexit`	Gate Exit	**Scan:** barcode, exit_image **Direct:** barcode:0, truck, is_own	Marks vehicle exited, logs duration
7	**GET** `boomexitedlist`	Gate Logs	company_id, from_date, to_date	Historical/today's exited vehicles list
8	**GET** `boomownvehiclelist`	Fleet	company_id	Registered company-owned vehicles
9	**POST** `boomownvehiclelog`	Fleet	truck, is_own:1, vehicle_type, [image]	Toggles entry/exit status for company fleet
10	**POST** `attendancelist`	HR	token, employee_id, company_id	Employee shift punch records
11	**POST** `getpayslip`	HR	token	List of issued payslip months & UUIDs
12	**POST** `publicpayslip`	HR	token, uuid	Binary PDF payslip document

5. Boom Barrier Hardware Controller Setup

The mobile app controls physical barrier gates by issuing local network POST requests to the controller switch IP:

- **Gate OPEN Command (Relay 2):** http://192.168.1.64:5000/relay/2/on — Triggered immediately after successful entry registration or barcode exit verification.
- **Gate CLOSE Command (Relay 1):** http://192.168.1.64:5000/relay/1/on — Triggered automatically after passage or via manual close override.
- **Dynamic IP Resolution:** The URLs are never hardcoded in the client. On login, the app requests /boomsettings which returns the exact configured IP endpoints for that specific quarry site.

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